The Oscillating Fragment

Student copy (project or hand out)

1 A rock no more than a hundred meters across, and perhaps less than half that, has kept Earth company for more than a century. Astronomers call it Kamo'oalewa. It orbits the Sun, not the Earth; but it holds so near our own path that, seen from here, it seems to circle us, a companion that never quite arrives and never quite departs. The technical name for such a body is a quasi-satellite: bound to the Sun, pacing the Earth, ours only in the loose sense of company. This one has shadowed us longer than the telescope that found it has existed, and by one orbital model it will go on doing so for a few hundred thousand years.

2 It was found on a single night in April 2016, by a survey telescope on Haleakala, in Hawaii, that sweeps the sky for things that move. The name came later, drawn from the Kumulipo, an old Hawaiian chant of creation: Ka mo'o a lewa, the oscillating fragment, a piece of something larger set swinging across the sky. The word fragment turned out to be a question rather than a description. A fragment of something, the name said, but of what it did not say.

3 For several years the likeliest answer pointed at the Moon. Kamo'oalewa reflects sunlight in a reddened way that resembles lunar soil left long exposed to space, and that resemblance fed an idea: that the rock is a piece of the Moon, thrown free by some ancient impact and never fully gone from Earth's neighborhood. A likely scar was even proposed, the crater Giordano Bruno on the Moon's far side, the right size and the right age to have flung such a fragment. The companion overhead, on this telling, was a relative.

4 Then the resemblance began to look less like proof. A group at the Chinese Academy of Sciences took ordinary material with no lunar history and weathered it by hand, firing lasers at it to mimic the long bombardment of open space. The color changed. It reddened, and went on reddening, until it matched Kamo'oalewa. The finding was narrow and sharp: a surface can turn that particular red simply by sitting in space long enough, lunar or not. The red that had pointed at the Moon now pointed only at time.

5 Into this uncertainty a spacecraft has arrived. Tianwen-2, launched from southwestern China in May of last year, reached Kamo'oalewa this past week and slipped into orbit around it, the first machine ever to keep company with Earth's companion. In the weeks ahead it will map the surface; in July it is to descend and take hold of the rock by two methods, one that anchors to the surface and one that only brushes it, and lift away at least a hundred grams of loose surface dust. That dust is due back on Earth in late 2027. A hundred grams is almost nothing, a small handful. It may settle the question the telescopes could not.

6 What it finds will not be the whole story. A hundred grams cannot speak for every rock that crosses our path, and a single sample can be read more than one way. But it will be the first piece of Kamo'oalewa held in a human hand rather than inferred from a smear of reflected light, and that is a different kind of

knowing. For now the fragment keeps its distance and its name, oscillating across the sky, a relative or a stranger, with the question, after more than a century, finally being put to the rock itself.

AP-style multiple choice:

1. The relationship between the third paragraph and the fourth paragraph is best described as one in which the writer first

A) proves that the asteroid came from the Moon, then explains how the returned sample will confirm it

B) dismisses the lunar origin as a myth, then offers a far-side crater as a better source

C) builds the case that the asteroid came from the Moon, then shows that its central evidence can be read another way

D) recounts the asteroid's discovery, then narrates the launch of the mission sent to reach it

2. In the fourth paragraph, the writer's use of the three-word sentence 'The color changed' serves chiefly to

A) summarize how the asteroid looks through the spacecraft's cameras

B) convey the writer's astonishment at how quickly the material reddened

C) mark the turn at which the experiment unsettles the lunar explanation, its bluntness lending the reversal weight

D) provide a neutral transition between two technical descriptions of the laboratory method

Discussion questions:

1. The writer waits until the fifth paragraph to bring in the spacecraft, after spending the first four paragraphs on what we do and do not know about the asteroid. How does that order change the way the mission reads when it finally appears?

2. The asteroid is called a 'companion,' then 'a relative,' and at the end 'a relative or a stranger.' What does the piece gain by leaving that family question open instead of resolving it?

Writing prompt (in-class or homework, Q2 rhetorical analysis):

Write an essay analyzing the rhetorical choices the writer uses to convey both wonder and uncertainty about the asteroid and the mission, without ever calling either one amazing. Consider elements such as the contrast between long and short sentences, word choice, the metaphor of kinship, and the careful handling of what is known and what is not.